POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 4041 | Semester IV | Theory | 4 Credits

Analog Communication

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 4041 as Analog Communication in Semester IV for Electronics Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage uses reliable public analog-communication curriculum sources and is not represented as recovered official SITTTR Course 4041 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders. The modules follow the cited public treatment of AM, DSB/SSB, FM/PM, noise, transmitters/receivers and pulse-modulation concepts; reconcile official details when available.

Verified source note: The official detailed source was inaccessible. Public sources supporting modulation, spectra, AM/FM generation and detection are linked in References.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Analog Communication explains how continuous-time information is transferred using a high-frequency carrier. It develops the concepts of modulation, spectra, bandwidth, generation/detection, noise, transmitters and receivers used in AM, SSB and angle-modulation systems.

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Prerequisite foundation

Sinusoids, frequency-domain ideas, basic electronic circuits and algebra/trigonometry.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain communication-system blocks and the need for modulation.
2. Analyse basic amplitude-modulation schemes and demodulation principles.
3. Explain FM/PM, bandwidth, generation/detection and comparison with AM.
4. Explain noise effects, receiver/transmitter blocks and multiplexing concepts.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO [illegible] exam-[illegible] [illegible] [illegible] [illegible] [illegible] [illegible]. Define, explain, apply [illegible] action words [illegible].

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain communication-system blocks, carrier modulation and basic AM spectrum concepts.	Understand
CO2	Compare AM, DSB-SC, SSB and VSB generation/detection approaches.	Apply
CO3	Explain angle modulation, frequency deviation, bandwidth and FM detection concepts.	Understand
CO4	Explain noise, receiver/transmitter blocks and basic FDM/TDM or pulse-modulation ideas.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Communication Systems and Amplitude Modulation	Communication blocks, message/carrier, need for modulation, AM waveform/spectrum/modulation index, power and envelope detection.	Approx. 15 hours	CO1	Module 1
2	DSB, SSB and VSB Systems	Suppressed-carrier and single-sideband systems, balanced/ring modulation, filtering/phase methods, coherent detection and bandwidth/power trade-offs.	Approx. 15 hours	CO2	Module 2
3	Angle Modulation and Noise	FM/PM, frequency deviation, modulation index, narrow/wideband, spectra/bandwidth, discriminators/PLL, noise and pre-emphasis/de-emphasis.	Approx. 16 hours	CO3	Module 3
4	Transmitters, Receivers and Multiplexing	AM/FM transmitter blocks, TRF/superheterodyne receivers, IF/AGC/limiting, pulse modulation and FDM/TDM.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Communication Systems and Amplitude Modulation

Communication blocks, message/carrier, need for modulation, AM waveform/spectrum/modulation index, power and envelope detection.

CO1 · Approx. 15 hours

DSB, SSB and VSB Systems

Suppressed-carrier and single-sideband systems, balanced/ring modulation, filtering/phase methods, coherent detection and bandwidth/power trade-offs.

CO2 · Approx. 15 hours

Angle Modulation and Noise

FM/PM, frequency deviation, modulation index, narrow/wideband, spectra/bandwidth, discriminators/PLL, noise and pre-emphasis/de-emphasis.

CO3 · Approx. 16 hours

Transmitters, Receivers and Multiplexing

AM/FM transmitter blocks, TRF/superheterodyne receivers, IF/AGC/limiting, pulse modulation and FDM/TDM.

CO4 · Approx. 14 hours

MODULE 1

Communication Systems and Amplitude Modulation

Communication blocks, message/carrier, need for modulation, AM waveform/spectrum/modulation index, power and envelope detection.

Approx. 15 hours | CO1 | Understand · Apply

Learning goals

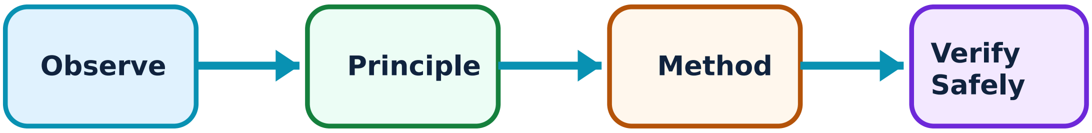
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Communication Systems and Amplitude Modulation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Why modulation is used−

A baseband signal is translated onto a higher-frequency carrier to enable practical antenna dimensions, channel allocation, propagation and simultaneous communication. Demodulation recovers the message at the receiver.

Study and application method

Draw source → transmitter/modulator → channel/noise → receiver/demodulator → destination and label message, carrier and modulated signal spectra.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw source → transmitter/modulator → channel/noise → receiver/demodulator → destination and label message, carrier and modulated signal spectra.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse the information signal with the carrier that transports it.

2. AM waveform and spectrum−

In AM, carrier amplitude varies with message amplitude. A single-tone AM signal has carrier and upper/lower sidebands; modulation index indicates depth and overmodulation causes distortion.

Study and application method

Sketch time-domain envelope and frequency components around carrier; state bandwidth and show how message frequency appears as sideband spacing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch time-domain envelope and frequency components around carrier; state bandwidth and show how message frequency appears as sideband spacing.

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Common mistake / precaution: Keep modulation index within the appropriate range for envelope detection; excessive modulation distorts the recovered message.

3. AM generation and detection−

AM can be generated using nonlinear/switching modulation arrangements and detected with envelope or synchronous techniques. Receiver filtering and component choices affect fidelity.

Study and application method

Identify carrier/message inputs, nonlinear/multiplying element, band-pass stage and detector/low-pass output path.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify carrier/message inputs, nonlinear/multiplying element, band-pass stage and detector/low-pass output path.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Detector time constants must suit carrier and message frequencies; arbitrary values create distortion or ripple.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

DSB, SSB and VSB Systems

Suppressed-carrier and single-sideband systems, balanced/ring modulation, filtering/phase methods, coherent detection and bandwidth/power trade-offs.

Approx. 15 hours | CO2 | Understand · Apply

Learning goals

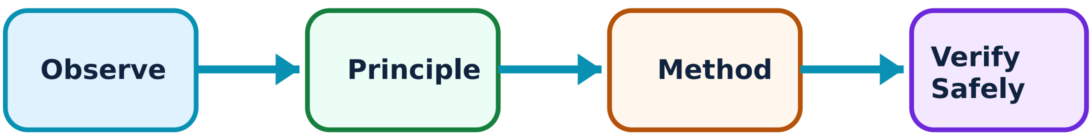
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for DSB, SSB and VSB Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. DSB-SC modulation−

DSB-SC suppresses the carrier and transmits both sidebands, improving carrier-power use but requiring coherent carrier recovery for demodulation.

Study and application method

Draw spectrum, identify absent carrier and explain why local oscillator phase/frequency alignment matters at the receiver.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw spectrum, identify absent carrier and explain why local oscillator phase/frequency alignment matters at the receiver.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A small carrier-frequency/phase mismatch can cause recovered-signal distortion in coherent detection.

2. SSB and VSB−

SSB retains one sideband to reduce bandwidth; VSB retains one full sideband and vestige of the other to balance filtering and low-frequency-message needs. Both are application-specific trade-offs.

Study and application method

Compare occupied bandwidth, carrier treatment, generation/detection complexity and common application context in a table.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare occupied bandwidth, carrier treatment, generation/detection complexity and common application context in a table.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not state SSB is always superior; receiver complexity and fidelity requirements matter.

3. Coherent detection−

Coherent detection multiplies the received signal by a locally generated carrier and filters the result. Carrier recovery/phase control is central to correct output.

Study and application method

Trace multiplier outputs and low-pass filtering, explaining desired baseband versus high-frequency product.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace multiplier outputs and low-pass filtering, explaining desired baseband versus high-frequency product.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not treat oscillator synchronisation as optional in suppressed-carrier systems.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Angle Modulation and Noise

FM/PM, frequency deviation, modulation index, narrow/wideband, spectra/bandwidth, discriminators/PLL, noise and pre-emphasis/de-emphasis.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

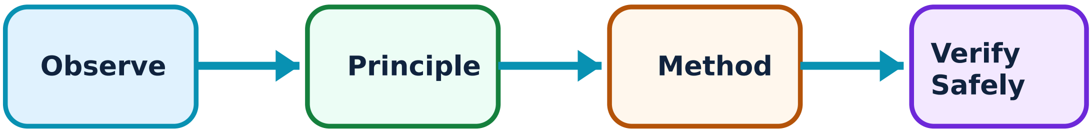
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Angle Modulation and Noise: identify the system, state its principle, follow the correct method and verify the result safely.

1. FM and PM principles−

FM changes carrier frequency and PM changes carrier phase in response to message. Both are angle modulation; frequency deviation and message frequency influence FM modulation index and spectrum.

Study and application method

Sketch carrier instantaneous frequency/phase change with message, label deviation and compare AM versus FM constant-amplitude behaviour.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch carrier instantaneous frequency/phase change with message, label deviation and compare AM versus FM constant-amplitude behaviour.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not confuse FM deviation with carrier frequency or assume FM has only two sidebands.

2. Bandwidth and detection−

FM produces multiple sidebands; practical occupied bandwidth is estimated using an approved rule and required performance. Frequency discriminators and phase-locked loops convert frequency/phase variation back to baseband.

Study and application method

List carrier, maximum message frequency and deviation; apply the approved bandwidth approach with units and identify detector blocks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List carrier, maximum message frequency and deviation; apply the approved bandwidth approach with units and identify detector blocks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Use only the formula/convention specified by the course or question; assumptions must be stated.

3. Noise and emphasis−

Noise degrades received signal quality differently across modulation schemes. Pre-emphasis boosts selected higher message frequencies before transmission; de-emphasis restores balance and improves perceived noise performance.

Study and application method

Identify where noise enters a system, state relevant SNR concept and draw pre-/de-emphasis placement in an FM chain.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify where noise enters a system, state relevant SNR concept and draw pre-/de-emphasis placement in an FM chain.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Do not claim a modulation scheme is immune to noise; compare conditions and threshold effects carefully.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Transmitters, Receivers and Multiplexing

AM/FM transmitter blocks, TRF/superheterodyne receivers, IF/AGC/limiting, pulse modulation and FDM/TDM.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

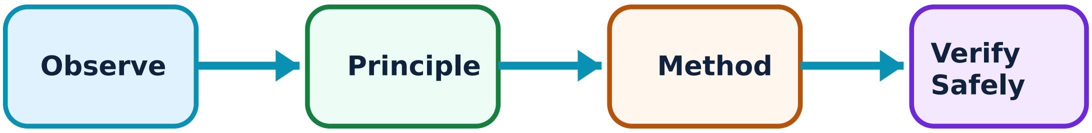
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Transmitters, Receivers and Multiplexing: identify the system, state its principle, follow the correct method and verify the result safely.

1. Transmitters and frequency stability−

Transmitters generate carrier, apply modulation, amplify and filter before radiation. Frequency stability and spectral control are essential to avoid interference.

Study and application method

Draw functional blocks and explain the role of oscillator, modulator, power amplifier, matching/filtering and antenna.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw functional blocks and explain the role of oscillator, modulator, power amplifier, matching/filtering and antenna.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: RF transmission is regulated; laboratory work must follow authorised low-power procedures and local rules.

2. Superheterodyne receiver−

A superheterodyne receiver mixes received RF with a local oscillator to a fixed intermediate frequency for amplification/selectivity, then detects and processes audio/baseband.

Study and application method

Trace RF filter → mixer/local oscillator → IF → detector → AF output, and explain why image rejection/tracking matter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace RF filter → mixer/local oscillator → IF → detector → AF output, and explain why image rejection/tracking matter.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□□ □□□□□□ □□□□□□□□□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Common mistake / precaution: Do not confuse IF with the received carrier frequency or local-oscillator frequency.

3. Pulse modulation and multiplexing−

PAM, PWM and PPM represent message samples through pulse amplitude, width or position. FDM separates channels by frequency; TDM assigns time slots, requiring timing/synchronisation.

Study and application method

Compare what parameter carries information, required bandwidth/timing and a suitable application; draw a simple frame/channel allocation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Sampling and timing limits must be respected; pulse concepts are not automatically equivalent to digital coding.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Communication Systems and Amplitude Modulation

Use the concept flow to revise observation, principle, method and verification.

Module 2: DSB, SSB and VSB Systems

Use the concept flow to revise observation, principle, method and verification.

Module 3: Angle Modulation and Noise

Use the concept flow to revise observation, principle, method and verification.

Module 4: Transmitters, Receivers and Multiplexing

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Draw spectra for a single-tone AM and a DSB-SC/SSB comparison.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Use simulation to compare AM envelope detection with coherent detection under instructor-approved conditions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Estimate FM occupied bandwidth for stated deviation/message limits using the prescribed rule.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Trace a superheterodyne receiver block diagram and label desired/undesired signals.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Communication Systems and Amplitude Modulation

Explain Why modulation is used and state one correct method, application or precaution.–

A baseband signal is translated onto a higher-frequency carrier to enable practical antenna dimensions, channel allocation, propagation and simultaneous communication. Demodulation recovers the message at the receiver.

Answer framework: Draw source → transmitter/modulator → channel/noise → receiver/demodulator → destination and label message, carrier and modulated signal spectra.

Important point: Do not confuse the information signal with the carrier that transports it.

Explain AM waveform and spectrum and state one correct method, application or precaution.−

In AM, carrier amplitude varies with message amplitude. A single-tone AM signal has carrier and upper/lower sidebands; modulation index indicates depth and overmodulation causes distortion.

Answer framework: Sketch time-domain envelope and frequency components around carrier; state bandwidth and show how message frequency appears as sideband spacing.

Important point: Keep modulation index within the appropriate range for envelope detection; excessive modulation distorts the recovered message.

Explain AM generation and detection and state one correct method, application or precaution.−

AM can be generated using nonlinear/switching modulation arrangements and detected with envelope or synchronous techniques. Receiver filtering and component choices affect fidelity.

Answer framework: Identify carrier/message inputs, nonlinear/multiplying element, band-pass stage and detector/low-pass output path.

Important point: Detector time constants must suit carrier and message frequencies; arbitrary values create distortion or ripple.

Module 2 — DSB, SSB and VSB Systems

Explain DSB-SC modulation and state one correct method, application or precaution.−

DSB-SC suppresses the carrier and transmits both sidebands, improving carrier-power use but requiring coherent carrier recovery for demodulation.

Answer framework: Draw spectrum, identify absent carrier and explain why local oscillator phase/frequency alignment matters at the receiver.

Important point: A small carrier-frequency/phase mismatch can cause recovered-signal distortion in coherent detection.

Explain SSB and VSB and state one correct method, application or precaution.−

SSB retains one sideband to reduce bandwidth; VSB retains one full sideband and vestige of the other to balance filtering and low-frequency-message needs. Both are application-specific trade-offs.

Answer framework: Compare occupied bandwidth, carrier treatment, generation/detection complexity and common application context in a table.

Important point: Do not state SSB is always superior; receiver complexity and fidelity requirements matter.

Explain Coherent detection and state one correct method, application or precaution.−

Coherent detection multiplies the received signal by a locally generated carrier and filters the result. Carrier recovery/phase control is central to correct output.

Answer framework: Trace multiplier outputs and low-pass filtering, explaining desired baseband versus high-frequency product.

Important point: Do not treat oscillator synchronisation as optional in suppressed-carrier systems.

Module 3 — Angle Modulation and Noise

Explain FM and PM principles and state one correct method, application or precaution.−

FM changes carrier frequency and PM changes carrier phase in response to message. Both are angle modulation; frequency deviation and message frequency influence FM modulation index and spectrum.

Answer framework: Sketch carrier instantaneous frequency/phase change with message, label deviation and compare AM versus FM constant-amplitude behaviour.

Important point: Do not confuse FM deviation with carrier frequency or assume FM has only two sidebands.

Explain Bandwidth and detection and state one correct method, application or precaution.−

FM produces multiple sidebands; practical occupied bandwidth is estimated using an approved rule and required performance. Frequency discriminators and phase-locked loops convert frequency/phase variation back to baseband.

Answer framework: List carrier, maximum message frequency and deviation; apply the approved bandwidth approach with units and identify detector blocks.

Important point: Use only the formula/convention specified by the course or question; assumptions must be stated.

Explain Noise and emphasis and state one correct method, application or precaution. −

Noise degrades received signal quality differently across modulation schemes. Pre-emphasis boosts selected higher message frequencies before transmission; de-emphasis restores balance and improves perceived noise performance.

Answer framework: Identify where noise enters a system, state relevant SNR concept and draw pre-/de-emphasis placement in an FM chain.

Important point: Do not claim a modulation scheme is immune to noise; compare conditions and threshold effects carefully.

Module 4 — Transmitters, Receivers and Multiplexing

Explain Transmitters and frequency stability and state one correct method, application or precaution.−

Transmitters generate carrier, apply modulation, amplify and filter before radiation. Frequency stability and spectral control are essential to avoid interference.

Answer framework: Draw functional blocks and explain the role of oscillator, modulator, power amplifier, matching/filtering and antenna.

Important point: RF transmission is regulated; laboratory work must follow authorised low-power procedures and local rules.

Explain Superheterodyne receiver and state one correct method, application or precaution.−

A superheterodyne receiver mixes received RF with a local oscillator to a fixed intermediate frequency for amplification/selectivity, then detects and processes audio/baseband.

Answer framework: Trace RF filter → mixer/local oscillator → IF → detector → AF output, and explain why image rejection/tracking matter.

Important point: Do not confuse IF with the received carrier frequency or local-oscillator frequency.

Explain Pulse modulation and multiplexing and state one correct method, application

or precaution.–

PAM, PWM and PPM represent message samples through pulse amplitude, width or position. FDM separates channels by frequency; TDM assigns time slots, requiring timing/synchronisation.

Answer framework: Compare what parameter carries information, required bandwidth/timing and a suitable application; draw a simple frame/channel allocation.

Important point: Sampling and timing limits must be respected; pulse concepts are not automatically equivalent to digital coding.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Communication Systems and Amplitude Modulation.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: DSB, SSB and VSB Systems.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Angle Modulation and Noise.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Transmitters, Receivers and Multiplexing.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Communication blocks, message/carrier, need for modulation, AM waveform/spectrum/modulation index, power and envelope detection..

2. Develop a complete answer or practical plan for: Suppressed-carrier and single-sideband systems, balanced/ring modulation, filtering/phase methods, coherent detection and bandwidth/power trade-offs..
3. Develop a complete answer or practical plan for: FM/PM, frequency deviation, modulation index, narrow/wideband, spectra/bandwidth, discriminators/PLL, noise and pre-emphasis/de-emphasis..
4. Develop a complete answer or practical plan for: AM/FM transmitter blocks, TRF/superheterodyne receivers, IF/AGC/limiting, pulse modulation and FDM/TDM..

LAST-MILE REVISION

Quick Revision

Module 1: Communication Systems and Amplitude Modulation

Communication blocks, message/carrier, need for modulation, AM waveform/spectrum/modulation index, power and envelope detection.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: DSB, SSB and VSB Systems

Suppressed-carrier and single-sideband systems, balanced/ring modulation, filtering/phase methods, coherent detection and bandwidth/power trade-offs.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Angle Modulation and Noise

FM/PM, frequency deviation, modulation index, narrow/wideband, spectra/bandwidth, discriminators/PLL, noise and pre-emphasis/de-emphasis.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Transmitters, Receivers and Multiplexing

AM/FM transmitter blocks, TRF/superheterodyne receivers, IF/AGC/limiting, pulse modulation and FDM/TDM.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Why modulation is used	A baseband signal is translated onto a higher-frequency carrier to enable practical antenna dimensions, channel allocation, propagation and simultaneous communication.
AM waveform and spectrum	In AM, carrier amplitude varies with message amplitude.
AM generation and detection	AM can be generated using nonlinear/switching modulation arrangements and detected with envelope or synchronous techniques.
DSB-SC modulation	DSB-SC suppresses the carrier and transmits both sidebands, improving carrier-power use but requiring coherent carrier recovery for demodulation.
SSB and VSB	SSB retains one sideband to reduce bandwidth; VSB retains one full sideband and vestige of the other to balance filtering and low-frequency-message needs.
Coherent detection	Coherent detection multiplies the received signal by a locally generated carrier and filters the result.
FM and PM principles	FM changes carrier frequency and PM changes carrier phase in response to message.
Bandwidth and detection	FM produces multiple sidebands; practical occupied bandwidth is estimated using an approved rule and required performance.
Noise and emphasis	Noise degrades received signal quality differently across modulation schemes.
Transmitters and frequency stability	Transmitters generate carrier, apply modulation, amplify and filter before radiation.
Superheterodyne receiver	A superheterodyne receiver mixes received RF with a local oscillator to a fixed intermediate frequency for amplification/selectivity, then detects and processes audio/baseband.
Pulse modulation and multiplexing	PAM, PWM and PPM represent message samples through pulse amplitude, width or position.

LEARNING RESOURCES

References

Recommended analog-communication references

1. B. P. Lathi, Communication Systems.
2. Simon Haykin, Communication Systems.
3. George Kennedy and Bernard Davis, Electronic Communication Systems.

Approved public analog-communication sources

- https://www.vignanlara.org/departments/syllabus/subjectwise/R13/ECE/RT22045.pdf
- https://www.cdt21.com/design_guide/analogue-modulation/

These sources support the study handbook while official Course 4041 content is unavailable; they do not replace an official detailed syllabus.